

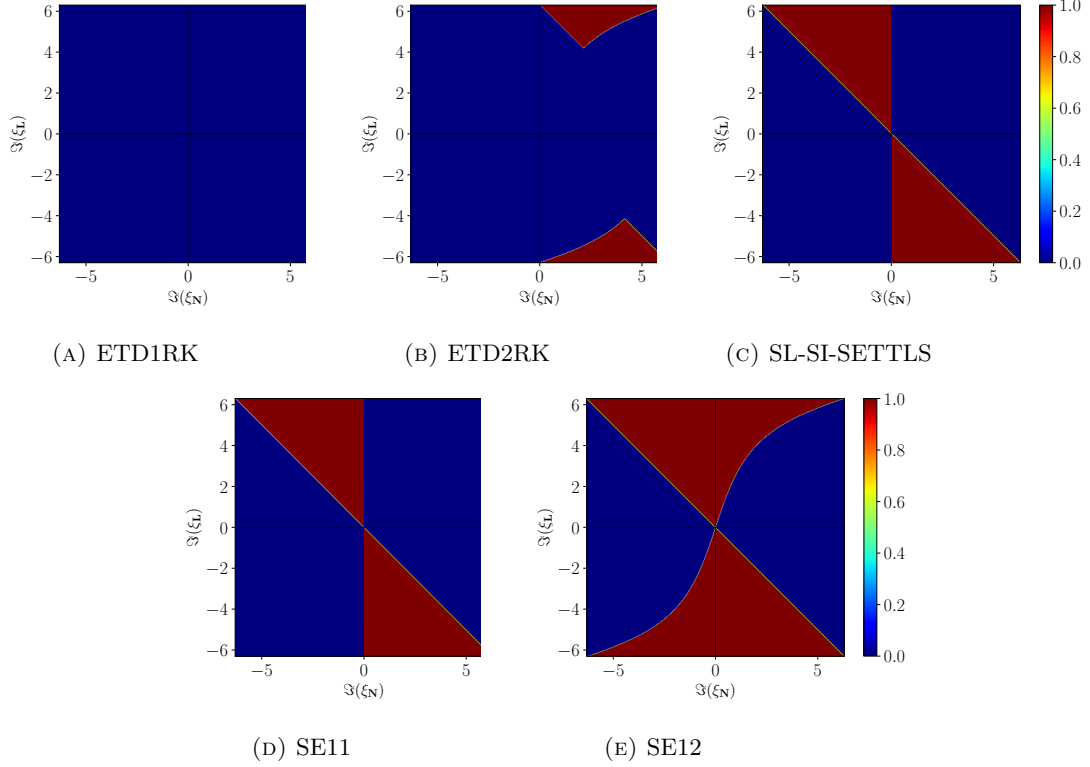


FIGURE 2. Stability regions (in red) of ETD1RK and ETD2RK applied to (6.4) (considering a nonlinear treatment of the advection term), and SL-SI-SETTLS, SE11 and SE12 applied to (6.3), in the $(\Im(\xi_N), \Im(\xi_L))$ plane, considering both ξ_N and ξ_L to be purely imaginary.

6.4. Stability of SE21 and SE22

Using the same procedure described above to assess stability properties of the proposed schemes SE21 and SE22 (Section 5) does not provide insights on potential stability improvements or decreases induced by the splitting of the matrix exponential. In particular, this approach is not able to capture the fact that the matrix exponential and the interpolation do not commute [26]. Indeed, using $u(x, t_n) = A^n e^{i\kappa x}$ in (5.7a) and (6.3), analogously to (6.9a), provides

$$\varphi_0\left(\frac{\xi_L}{2}\right) e^{-i\kappa s} \varphi_0\left(\frac{\xi_L}{2}\right) = e^{\xi_L/2} e^{-i\kappa s} e^{\xi_L/2} = e^{\xi_L} e^{-i\kappa s} = \varphi_0(\xi_L) e^{-i\kappa s} \quad (6.10)$$

such that $A_{SE21} \equiv A_{SE11}$, and, similarly, $A_{SE22} \equiv A_{SE12}$. However, different stability properties are verified in the numerical simulations presented in Section 7.

Next, we develop in detail how the differences between the two sets of schemes arise. Consider the ODE (6.3) with only the linear term:

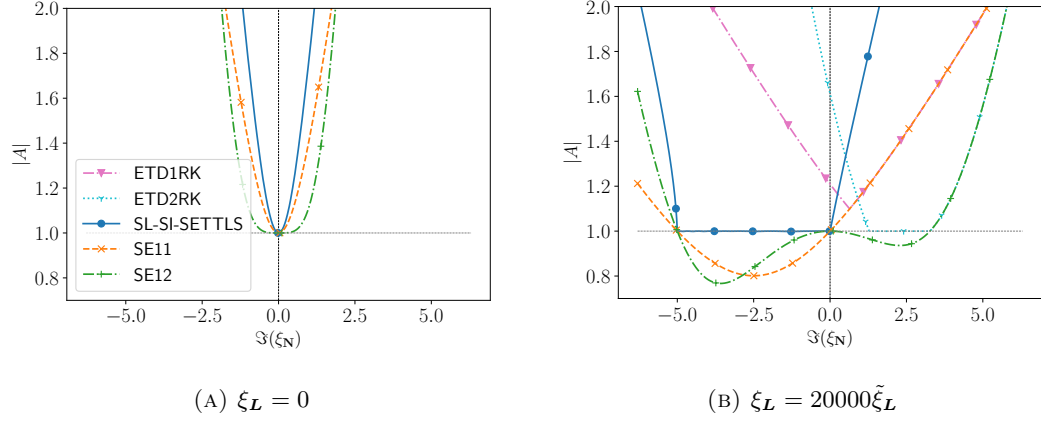


FIGURE 3. Stability functions of ETD1RK and ETD2RK applied to (6.4) (considering a nonlinear treatment of the advection term), and SL-SI-SETTLS, SE11 and SE12 applied to (6.3), as a function of $\Im(\xi_N)$ for fixed values of ξ_L and $\Re(\xi_N) = 0$. The stability regions of ETD1RK and ETD2RK are empty in the case $\xi_L = 0$. The horizontal, dashed line indicates the boundary of the stability region requiring $|A| \leq 1$. The same legend holds for both plots.

$$\frac{du}{dt} = \lambda_L u \quad (6.11)$$

We first consider the first-order discretization used in (3.15a). Since the problem is solved in the spectral space, we may write

$$\hat{u}_k^{n+1} = \varphi_0(\xi_{L_k})[\hat{u}_k^n]_*, \quad k = -M, \dots, M \quad (6.12)$$

where $u(t, x) \approx \sum_{k=-M}^M \hat{u}_k^n e^{i\kappa x}$ is the truncated Fourier decomposition of u and $\xi_{L_k} := \Delta t \lambda_{L_k}$ may depend on k . Note, however, that the interpolation is performed on the grid. Therefore, using the direct and inverse discrete Fourier transforms

$$\hat{u}_k^n = \mathcal{F}(u^n) = \frac{1}{P} \sum_{j=0}^{P-1} u(t_n, x_j) e^{-i\kappa x_j}, \quad u(t_n, x_j) = \mathcal{F}^{-1}(\hat{u}^n) = \sum_{k=-M}^M \hat{u}_k^n e^{i\kappa x_j} \quad (6.13)$$

with $\kappa = 2\pi k/L$, L the length of the periodic spatial domain and $P+1$ the number of equidistant grid points $x_i = i\Delta x$, $i = 0, \dots, P$, $\Delta x = L/P$, we may rewrite (6.12) more rigorously as

$$\begin{aligned} \hat{u}_k^{n+1} &= \varphi_0(\xi_{L_k}) \mathcal{F}([\mathcal{F}^{-1}(\hat{u}^n)]_*) = \varphi_0(\xi_{L_k}) \mathcal{F}([u_j^n]_*) = \varphi_0(\xi_{L_k}) \frac{1}{P} \sum_{j=0}^{P-1} u(t_n, x_{d_j}) e^{-i\kappa x_j} \\ &= \varphi_0(\xi_{L_k}) \frac{1}{P} \sum_{j=0}^{P-1} u(t_n, x_{j-p_j}) e^{-i\kappa x_j} = \varphi_0(\xi_{L_k}) \frac{1}{P} \sum_{j=0}^{P-1} u(t_n, x_{r_j}) e^{-i\kappa x_{p_j+r_j}} \end{aligned}$$

$$= \varphi_0(\xi_{\mathbf{L}_k}) \frac{1}{P} \sum_{j=0}^{P-1} u(t_n, x_{r_j}) e^{-i\kappa(p_j+r_j)\Delta x}, \quad (6.14)$$

where x_{d_j} is the approximate departure point at t_n of the Lagrangian trajectory arriving at x_j at t_{n+1} , $r_j := j - p_j$, and, for simplification, $x_{d_j} = x_{j-p_j}$, $p_j \in \mathbb{Z}$, is supposed to belong to the grid (otherwise $u(t_n, x_{d_j})$ may be written as a linear combination (*i.e.*, interpolation) of the solution computed at grid points). If the velocity field is constant in space, then $\forall j, p_j = p$, thus

$$\begin{aligned} \hat{u}_k^{n+1} &= \varphi_0(\xi_{\mathbf{L}_k}) e^{-i\kappa p \Delta x} \frac{1}{P} \sum_{j=0}^{P-1} u(t_n, x_{r_j}) e^{-i\kappa x_{r_j}} \\ &= \varphi_0(\xi_{\mathbf{L}_k}) e^{-i\kappa p \Delta x} \frac{1}{P} \sum_{r=0}^{P-1} u(t_n, x_r) e^{-i\kappa x_r} = \varphi_0(\xi_{\mathbf{L}_k}) \hat{u}_k^n e^{-i\kappa p \Delta x} = \varphi_0(\xi_{\mathbf{L}_k}) \hat{u}_k^n e^{-i\kappa s}, \end{aligned} \quad (6.15)$$

with $s := x_j - x_{d_j} = x_j - x_{j-p} = p\Delta x$. Therefore, for each k , the stability function $|\hat{u}_k^{n+1}|/|\hat{u}_k^n| = \varphi_0(\xi_{\mathbf{L}_k}) e^{-i\kappa s}$ is the same as derived above (see Eq. (6.9a)), which corresponds, in fact, the exact solution for wavenumber k . However, if the velocity field is not constant, we are not able to factor out $e^{-i\kappa p_j \Delta x}$ from the summation, and the stability function may be different, and, in particular, the exact solution for wavenumber k is no longer obtained; this suggests an intricate relation between spectral transforms, matrix exponentiation, and spatial interpolation.

Consider now the proposed modification to the discretization of the linear term. In the spectral space, we solve

$$\hat{u}_k^{n+1} = \varphi_0\left(\frac{\xi_{\mathbf{L}_k}}{2}\right) \mathcal{F}([\mathcal{F}^{-1}(\hat{w})^n]_*), \quad k = -M, \dots, M \quad (6.16)$$

where

$$\hat{w}_k^n := \varphi_0\left(\frac{\xi_{\mathbf{L}_k}}{2}\right) \hat{u}_k^n. \quad (6.17)$$

Conducting the same development as above,

$$\hat{u}_k^{n+1} = \varphi_0\left(\frac{\xi_{\mathbf{L}_k}}{2}\right) \frac{1}{P} \sum_{j=0}^{P-1} w(t_n, x_{r_j}) e^{-i\kappa(p_j+r_j)\Delta x} = \varphi_0\left(\frac{\xi_{\mathbf{L}_k}}{2}\right) \hat{w}_k^n e^{-i\kappa s} = \varphi_0(\xi_{\mathbf{L}_k}) \hat{u}_k^n e^{-i\kappa s}, \quad (6.18)$$

where the second equality holds if the velocity field is constant. We conclude that, if v is constant, the original and the proposed discretizations of the linear term yield the same stability function. However, if the velocity field is not constant, we have

$$\begin{aligned} \hat{u}_k^{n+1} &= \varphi_0\left(\frac{\xi_{\mathbf{L}_k}}{2}\right) \frac{1}{P} \sum_{j=0}^{P-1} w(t_n, x_{r_j}) e^{-i\kappa(p_j+r_j)\Delta x} \\ &= \varphi_0\left(\frac{\xi_{\mathbf{L}_k}}{2}\right) \frac{1}{P} \sum_{j=0}^{P-1} \left[\sum_{l=-M}^M \hat{w}_l^n e^{i\kappa_l x_{r_j}} \right] e^{-i\kappa(p_j+r_j)\Delta x} \end{aligned}$$